

# SteamScope: A Database-Driven Game Discovery and Analytics Platform

## 01 Abstract

SteamScope is a database-driven web application designed to help players discover games they may enjoy while also exploring the data behind the Steam catalog. The project begins with a real Steam catalog containing 141,899 imported entries. During preparation, an unreliable CSV source was investigated and replaced with a clean JSON source, which was streamed, cleaned, and transformed into structured game and relationship data. The final MySQL design contains 24 related tables, including reusable tables for developers, publishers, genres, tags, categories, platforms, and languages. Junction tables represent games with multiple attributes, while primary keys and foreign keys maintain valid links between records. The database also includes a simulated community of 3,000 profiles with libraries, wishlists, favorites, purchases, achievements, and play activity. A FastAPI backend uses parameterized SQL to provide browsing, filtering, analytics, rankings, collection management, and explainable recommendations. Recommendations compare genres, tags, and developers from a profile's library, giving additional weight to favorite games. A React frontend presents these features through catalog discovery, personal collections, rankings, analytics, and an interactive database presentation. SteamScope demonstrates how careful data cleaning, relational database design, SQL queries, and a web interface can work together to turn a large catalog into useful and understandable game discovery.

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Keywords: Steam catalog, relational database, MySQL, FastAPI, React, game discovery, analytics, recommendations, data cleaning